

This instruction manual is divided into two parts: Chinese and English. The first part is in Chinese, and the second part is in English.

This instruction is divided into two parts, Chinese and English. The Chinese description is in the front and the English description is at the back.

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1. Real-time data

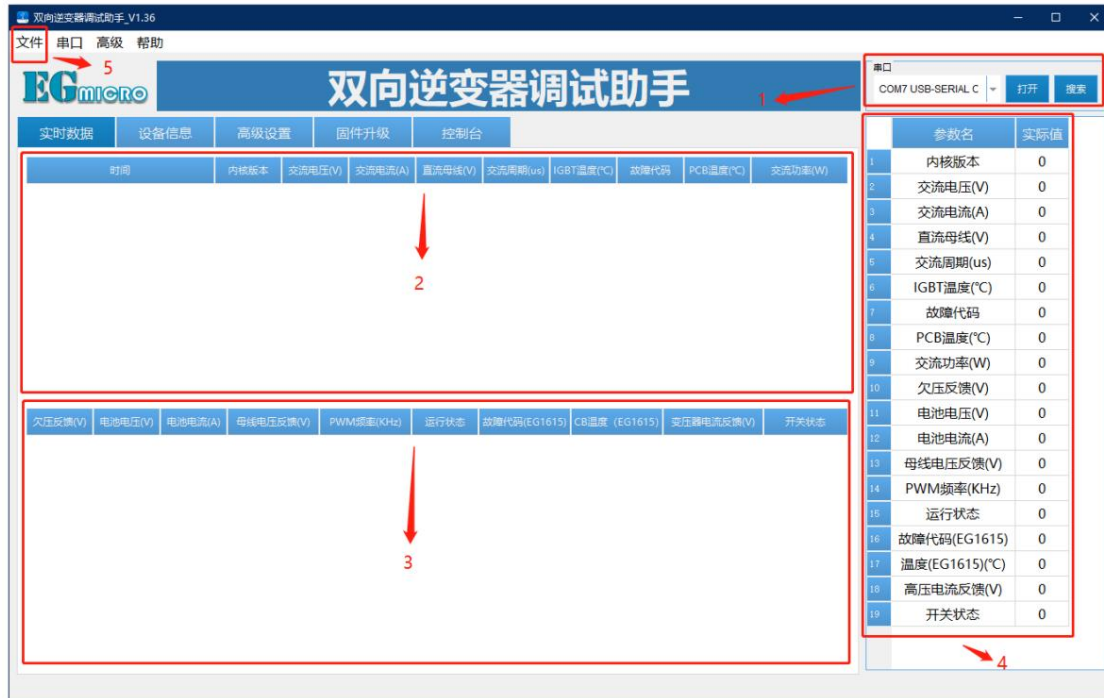


Figure 1-1

1. Connect to serial port: After clicking search, select the CH340 device, then click open.

Inverter section data display 3.

Battery section data display 4. All

data display: Temperature >40°C or <0°C will be highlighted in red; abnormal fault codes will also be

highlighted in red. 5. Export data: Open the "File" button, select "Export to Excel" to export the data from steps 2 and 3; if you encounter any testing problems, send the exported data and the problem symptoms to an engineer for analysis.

2 Equipment Information



Figure 2-1

1. Read DID Configuration: Click "Read" to read a single DID configuration, or click "Read All" to read all DID configurations. 2. Enter the value to be written. 3.

Write DID Configuration: Click "Write" to write the value to be set; the setting takes effect

immediately. 4. Open-loop output, closed-loop output: Used for low-voltage I/O testing; prohibited during normal operation.

3 Advanced Settings

3.1 EG1615 Parameter Modification



Figure 3-1-1

1. Total Battery Setting and SOC: Reserved. 2. Read CFG

Configuration: Click "Read" to read a single CFG configuration entry; click "Read All" to read all CFG configuration entries. 3. Write CFG Configuration: Click "Write" to write the value

to be set in the corresponding register. Ensure parameter integrity; this operation does not guarantee accuracy.

It will be stored and restored if the power is lost.

4. Save All: Clicking the "Save All" button will save all parameters. 5. Restore Defaults: Clicking the "Restore Defaults" button

will restore advanced settings to factory defaults; device information parameters will not be restored.

complex.

PS: After restoring to default settings, wait 5 seconds before powering on again.

3.2 EG8026/EG8025 Parameter Modification



Figure 3-2-1



Figure 3-2-2

PS: After performing steps 1-4 below on the EG8026, you need to set the EG1615 Advanced Settings - Inverter Control Parameters to 0 (i.e., custom mode) for the modified parameters to work. For the EG8025, simply perform steps 1-4 below.

1. Read CFG Configuration: As shown in Figure 3-2-1, click "Read All" to read all CFG configurations. 2. Save All: As shown in Figure 3-2-1, after clicking the "Save All" button, all parameters will be saved to the custom area. 3. Switch to Custom Area: As shown in Figure 3-2-2, enter 0 in the "Inverter Configuration to be Set" field and click "Write" to switch to the custom area.
- Custom zone. PS: Inverter configuration 0 is the custom zone, 1 is the default parameter.
4. Modify parameters: As shown in Figure 3-2-1, click "Write" to write the value to be set in the corresponding register. Ensure the parameter is complete. This operation is not saved and will be restored after power failure. The modified parameters will be saved after clicking the "Save All" button.
5. Restore Defaults: As shown in Figure 3-2-1, click the "Restore Defaults" button. All information, including the device settings, will be restored to factory defaults.
- Information parameters.
- PS: After restoring to default settings, wait 5 seconds before powering on again.

4. Firmware upgrade

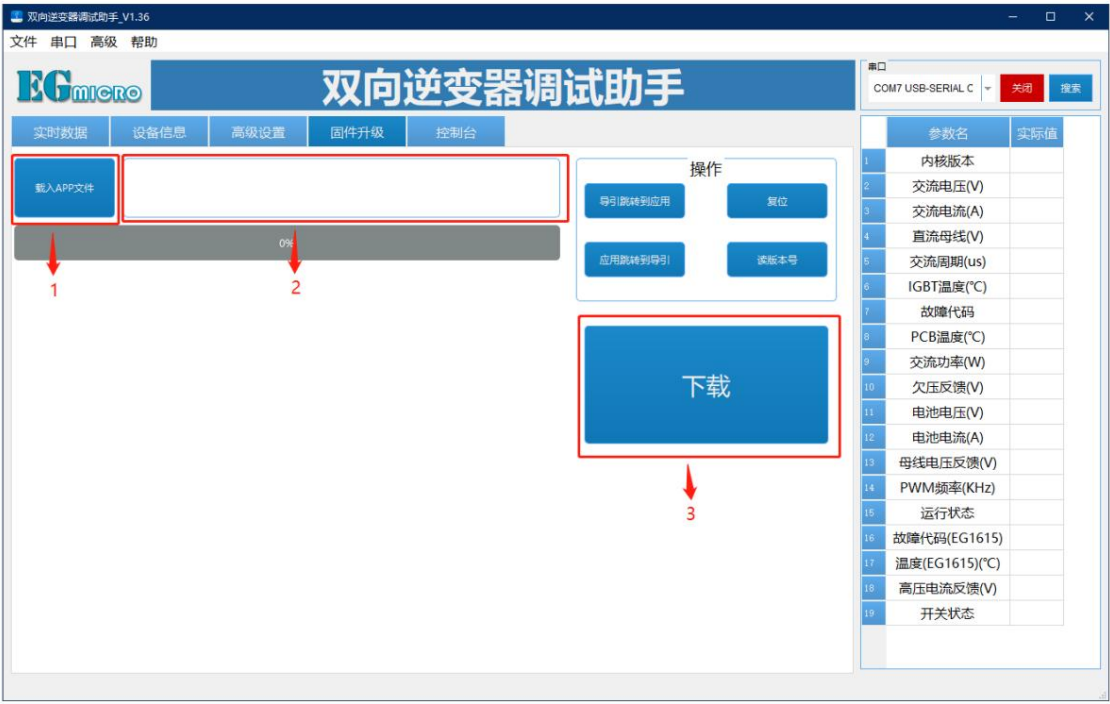


Figure 4-1

1. Load App File: Click "Load App File" and select the file. 2. Display File Path: Files can also be dragged and dropped directly. 3. Update Program: Click the "Download" button to enter the program download mode.
- PS
If a download error occurs, simply click the "Download" button again. If the serial port is connected to the EG1615 while the device is in operation, and you need to update the EG8026 program, you must first pass the data from the EG1615 to the EG8026, then click download. After the download is complete, you need to pass the data back.
- EG1615. (Please refer to the console instructions for the pass-through function.)
- If updating the driver board program separately, if the serial port is connected to the EG1615, directly load the EG1615's APP file; if the serial port is connected to the EG8026, directly load the EG8026's APP file.

5. Console

5.1 EG1615 Console

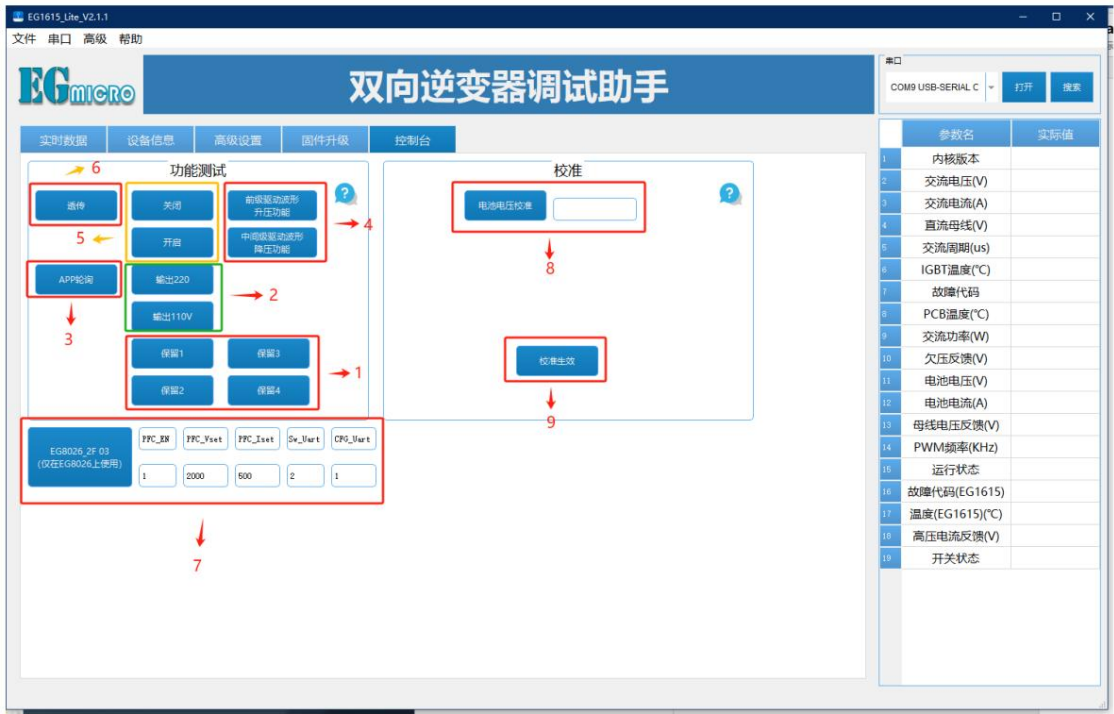


Figure 5-1

PS. Instructions for using the serial port on the EG1615: 1. The four

buttons are for debugging purposes and have no actual function. 2. 220V

output: Switch the EG8026 to configuration 1; 110V output: Switch the EG8026 to configuration 2. Actual output will vary depending on the specific configuration.

The configuration of EG8026 shall prevail.

3. APP Polling: When session is 3, APP messages stop. APP data can be retrieved via this button. 4. Boost Test, Buck Test: This button is for testing purposes only and

should not be used during normal operation. 5. Enable: Discharge mode enables the inverter, charging mode enables charging; Disable: Discharge mode disables the inverter, charging

mode stops charging. 6. Pass-through: This means receiving and operating commands from the EG8026 via the

EG1615.

Specific operation: When the serial port is connected to the EG1615, clicking the "Pass Through" button allows you to read the device information and advanced settings parameters

of the EG8026. You can also use the EG8026's calibration and parameter modification functions. Clicking the "Pass Through" button again switches back to the EG1615. Under

normal operating conditions, the serial port needs to read data from the EG1615; otherwise, the inverter may not output data or the charger may not charge.

7. EG8026_2F_03 Button: This button is for testing the PFC function of the EG8026. It should not be used during normal operation. 8. Calibrate Battery Voltage: Under

normal operating conditions, measure the actual battery voltage, multiply by 100, retain the integer part, and enter it into the input box on the right. Then click the "Calibrate Battery Voltage"

button. The calibration is not saved at this time.

9. Calibration in Effect: After ensuring the calibration is correct, click the "Calibration in Effect" button. The calibration value will then be saved.

PS: All calibrations must be performed in inverter mode.

5.2 EG8026 Control Panel



Figure 5-2

5.2.1 EG8026 Console User Manual

PS Instructions for using the serial port with the EG8026 (All calibrations must be performed in inverter mode, and current calibration must be performed under full load).

1. Closed-loop output: Controls the closed-loop output of EG8026; Open-loop output: Controls the open-loop output of EG8026. 2. Output voltage calibration: Under normal operating conditions, measure the actual AC voltage value, multiply by 10, retain the integer part, and enter it into the input box on the right. Then click the "Output Voltage Calibration" button. Calibration is not saved at this time. 3. Bus voltage calibration: Under normal operating conditions, measure the actual bus voltage value, multiply by 10, retain the integer part, and enter it into the input box on the right. Then click the "Bus Voltage Calibration" button. Calibration is not saved at this time. 4. Output current calibration: (This calibration value corresponds to the total current of the sampling resistor under the H-bridge arm) Under normal operating conditions, measure the actual AC current value, multiply by 100, retain the integer part, and enter it into the input box on the right. Then click the "Output Current Calibration" button. Calibration is not saved at this time. 5. Hall Current Calibration: (This calibration value corresponds to the current acquired by the Hall chip) Under normal operating conditions, measure the actual AC current, multiply by 100, retain the integer part, and enter it into the input box on the right. Then click the "Hall Current Calibration" button. The calibration is not saved at this time. 6. Calibration Activation: After ensuring the calibration is correct, click the "Calibration Activation" button. All calibration values will then be saved.

P.S. For the other buttons, please refer to the [EG1615 console documentation](#).

5.2.1 EG8025 Console User Manual

PS2. Instructions for using the serial port with the EG8025 (all calibrations must be performed in inverter mode, and current calibration must be performed under full load): 1. Closed-loop output: Controls the closed-loop output of the EG8025; Open-loop output: Controls the open-loop output of the EG8025; 2. Output voltage calibration: Under normal operating conditions, measure the actual AC voltage value, multiply it by 10, retain the integer part, and enter it into the input box on the right. Then click the "Output Voltage Calibration" button. Calibration is not saved at this time. 3. Bus voltage calibration: Under normal operating conditions, measure the actual bus voltage value, multiply it by 10, retain the integer part, and enter it into the input box on the right. Then click the "Bus Voltage Calibration" button. Calibration is not saved at this time. 4. Output current calibration (this calibration value corresponds to the sampling resistor current of the left arm of the H-bridge in the EG8025): This calibration method is not recommended for the EG8025. 5. Hall current calibration (this calibration value corresponds to the sampling resistor current of the right arm of the H-bridge in the EG8025): This calibration method is not recommended for the EG8025. 6. Calibration effective: After ensuring the calibration is correct, click the "Calibration effective" button. At this time, all calibration values will be saved.

P.S. For the other buttons, please refer to the EG1615 console documentation.

6 Real-Time Data

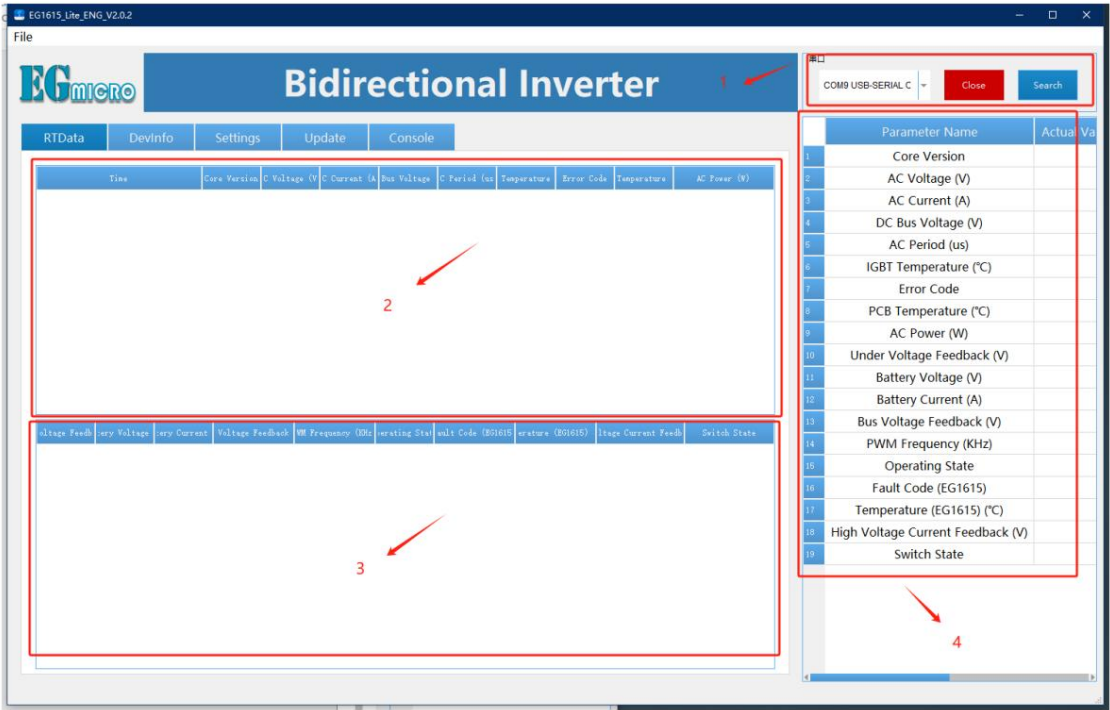


Figure 6-1

- 1. **Connect the serial port:** After clicking "Search", select the CH340 device, and then click "Open".
- 2. **Inverter part data display**

3. **Battery part data display**
4. **All data display:** Temperatures $> 40^{\circ}\text{C}$ or $\leq 0^{\circ}\text{C}$ will be marked red, and abnormal fault codes will be marked red.
5. **Export data:** Click the "File" button and select "Export to Excel" to export the data in items 2 and 3. In case of testing problems, send the exported data and the problem phenomena to the engineer for analysis.

7 Device Information

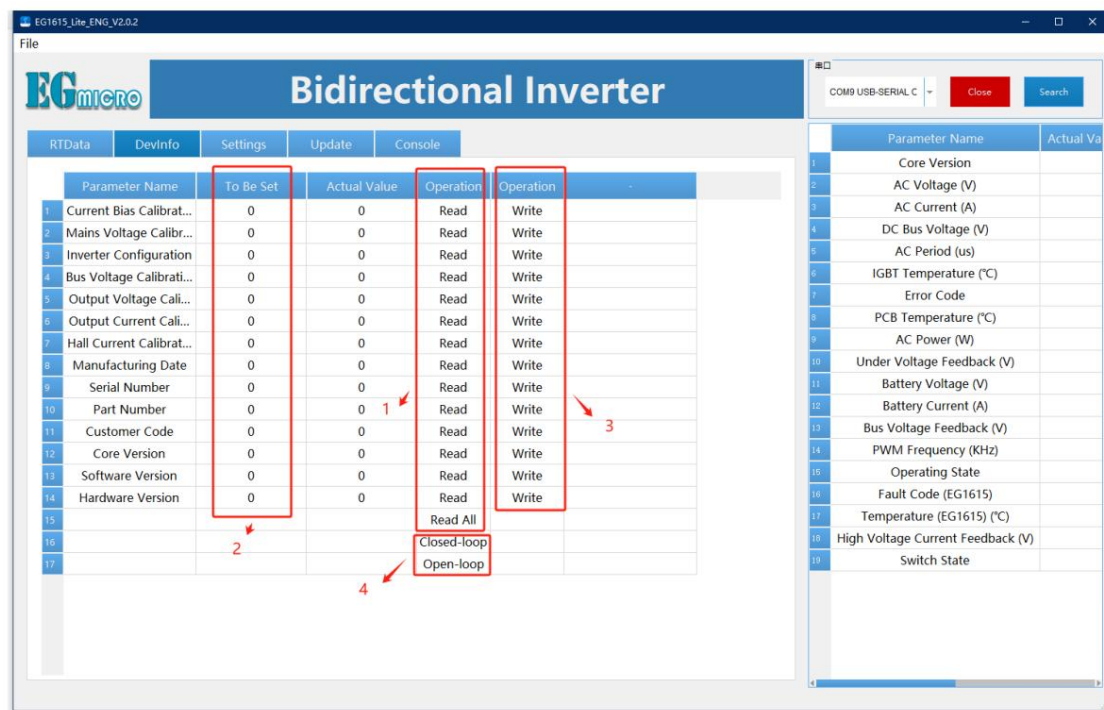


Figure 7-1

1. **Read DID configuration:** Click "Read" to read a single DID configuration, and click "Read All" to read all DID configurations.
2. Enter the value to be written.
3. **Write DID configuration:** Click "Write" to write the value to be set, and the setting takes effect immediately upon writing.
4. **Open-loop output, closed-loop output:** For low-voltage IO testing. Prohibited during normal operation.

8 Setting

8.1 EG1615 Parameter Modification

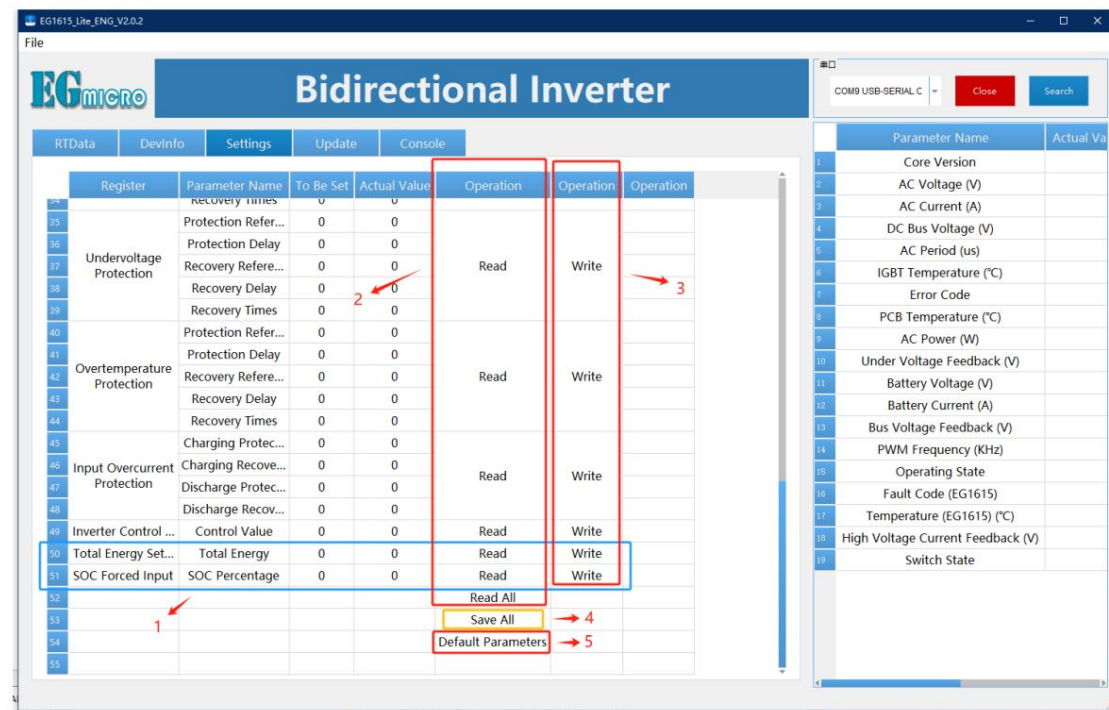


Figure 8-1-1

1. **Total Energy Setting and SOC Forced Input:** Reserved
 2. **Read CFG configuration:** Click "Read" to read a single CFG configuration, and click "Read All" to read all CFG configurations.
 3. **Write CFG configuration:** Click "Write" to write the value to be set in the corresponding register. Ensure the parameters are complete. This operation is not saved, and the settings will be restored after power loss.
 4. **Save all:** After clicking the "Save All" button, all parameters will be saved.
 5. **Restore defaults:** After clicking the "Restore Defaults" button, the advanced setting parameters will be restored to the factory settings, while the device information parameters will not be restored.
- PS: Wait for 5 seconds after restoring defaults and then power on again.

8.2 EG8026/EG8025 Parameter Modification

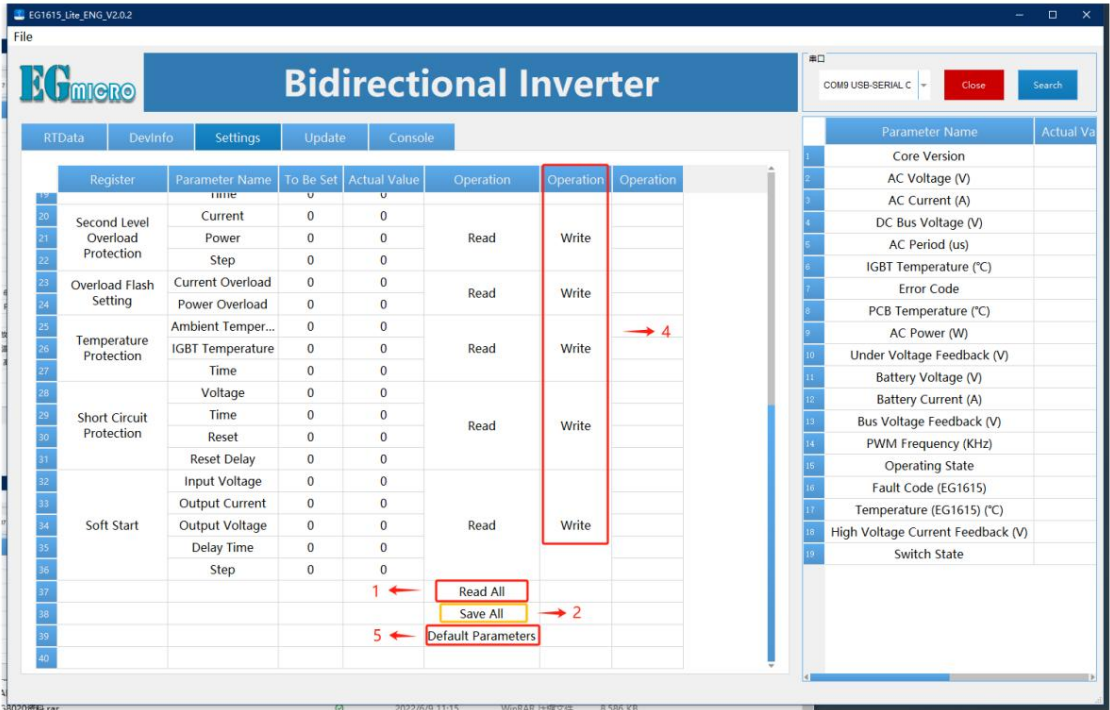


Figure 8-2-1

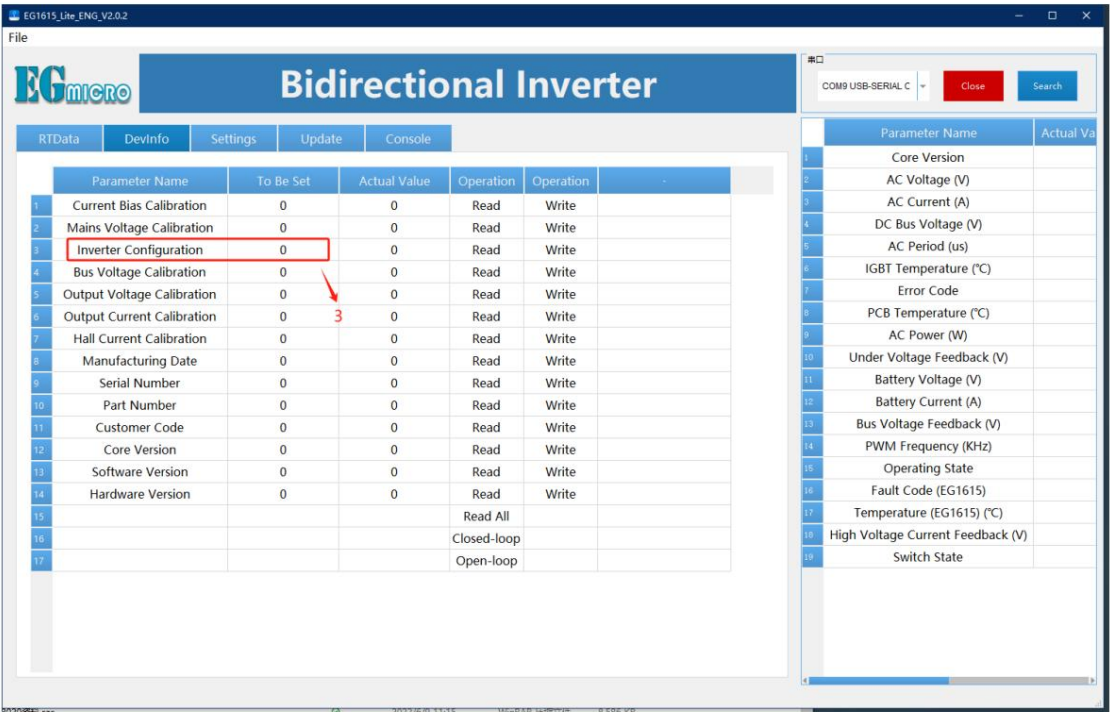


Figure 8-2-2

PS: After performing steps 1 - 4 below for EG8026, the inverter control parameters in the advanced settings of EG1615 need to be written as 0 (i.e., custom mode) for the modified parameters to take effect. For EG8025, just perform steps 1 - 4 below.

1. **Read CFG configuration:** As shown in Figure 8 - 2 - 1, click "Read All" to read all CFG configurations.
2. **Save all:** As shown in Figure 8 - 2 - 1, after clicking the "Save All" button, save all parameters to the custom area.
3. **Switch to the custom area:** As shown in Figure 8 - 2 - 2, fill in 0 for the inverter configuration to be set, and then click "Write" to switch to the custom area. PS: Inverter configuration 0 represents the custom area, and 1 represents the default parameters.
4. **Modify parameters:** As shown in Figure 8 - 2 - 1, click "Write" to write the value to be set in the corresponding register. Ensure the parameters are complete. This operation is not saved, and the settings will be restored after power loss. Only after clicking the "Save All" button will the modified parameters be saved.
5. **Restore defaults:** As shown in Figure 8 - 2 - 1, click the "Restore Defaults" button, and all information will be restored to the factory settings, including the parameters of the device information.

PS: Wait for 5 seconds after restoring defaults and then power on again.

9 Update

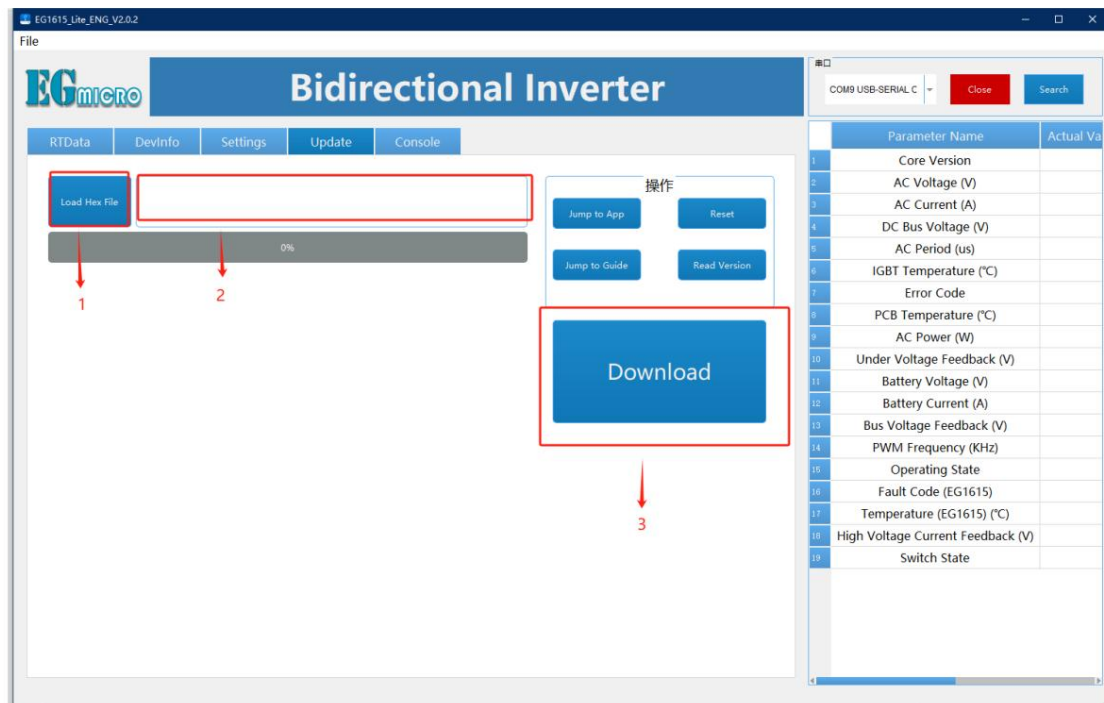


Figure 9-1

1. **Load APP file:** Click "Load APP File" and select the file.
2. **Display file path:** Files can also be directly dragged in.
3. **Update the program:** Click the "Download" button to enter the program download mode.

PS:

ÿ In case of a download error, simply click the "Download" button again.

ÿ When the whole machine is in operation and the serial port is connected to EG1615, if you need to update the program of EG8026, you need to pass through from EG1615 to EG8026 first, and then click to download. After the download is completed, it needs to be passed back to EG1615.

(For the pass-through function, please refer to the console instructions.)

ÿ If you are updating the driver board program alone, when the serial port is connected to EG1615, directly load the APP file of EG1615. When the serial port is connected to EG8026, directly load the APP file of EG8026.

10 Console

10.1 EG1615 Console

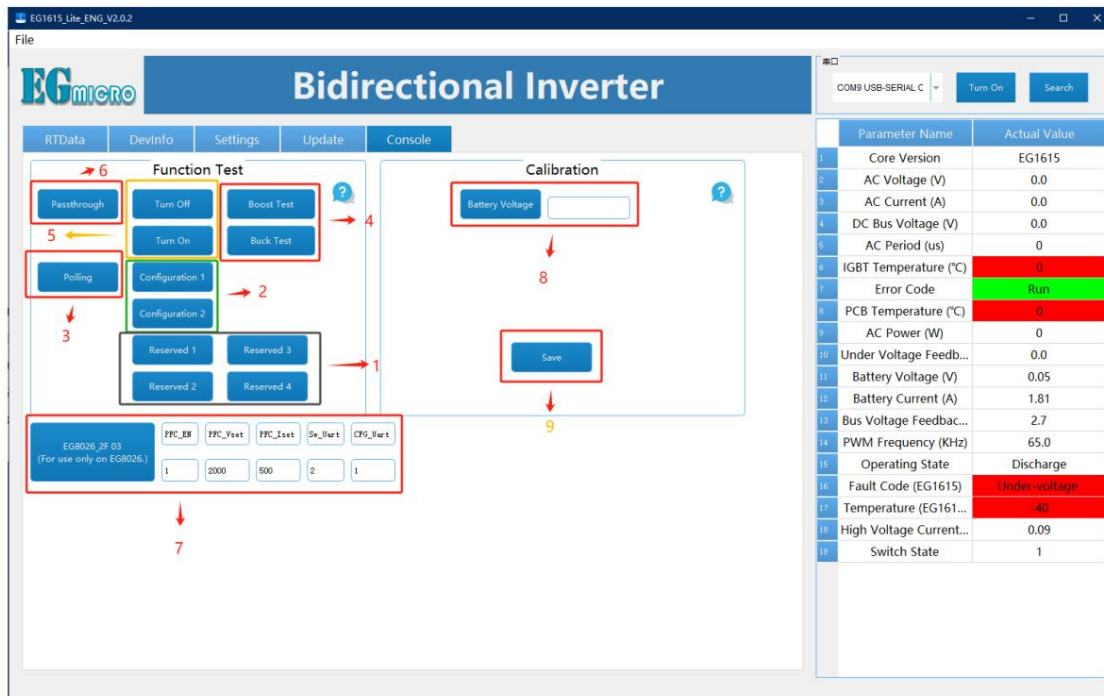


Figure 10-1-1

PS. Instructions for Using When the Serial Port is Connected to EG1615

1. The four buttons are for debugging purposes and have no practical functions.
2. **Configuration 1:** Switch EG8026 to Configuration 1; **Configuration 2:** Switch EG8026 to Configuration 2. The actual output voltage is subject to the configuration of EG8026.
3. **Polling:** When the session is 3, the APP message stops. You can obtain APP data through this button.
4. **Boost Test, Buck Test:** These buttons are for testing purposes only and are used only in low-voltage IO testing. They are prohibited during normal operation.
5. **Turn On:** In the discharging state, it turns on the inverter; in the charging state, it starts charging.
Turn Off: In the discharging state, it turns off the inverter; in the charging state, it stops charging.
6. **Passthrough:** It means receiving and operating the instructions of EG8026 through EG1615.
Specific operation: When the serial port is connected to EG1615, after clicking the "Passthrough" button, you can read the device information and advanced setting parameters of EG8026. At the same time, you can also use functions such as calibration and parameter modification of EG8026. After clicking the "Passthrough" button again, you can switch back to EG1615. During normal operation, the serial port needs to read the data of EG1615. Otherwise, there may be situations where the inverter does not output or the charging does not work.
7. **EG8026_2F_03 Button:** This button is for testing the PFC function of EG8026. It is prohibited during normal operation.
8. **Calibrate Battery Voltage:** During normal operation, measure the actual battery voltage value, multiply it by 100, retain the integer part, fill it into the input box on the right, and then click the " Battery Voltage" button. At this time, the calibration is not saved.
9. **Calibration Save:** After ensuring the calibration is correct, click the "Save" button. At this time, the calibration value is saved.

PS: All calibrations need to be carried out in the inverter state.

10.2·EG8026 Console

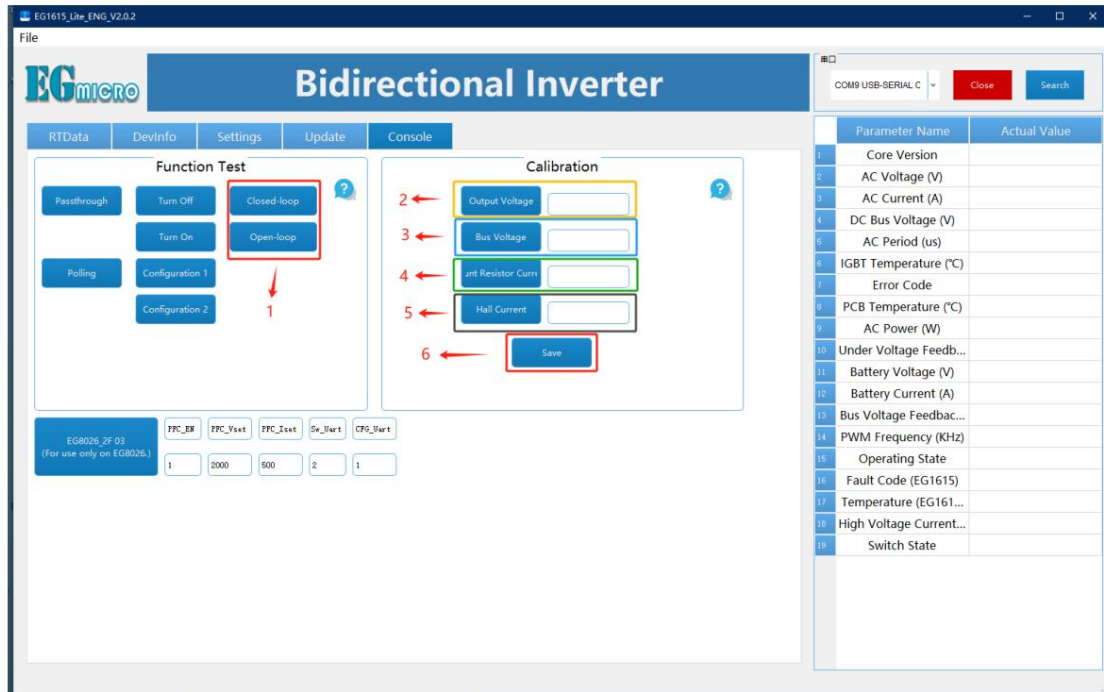


Figure 10-2-1

10.2.1·EG8026 Console Usage Instructions

PS. Instructions for Using When the Serial Port is Connected to EG8026 (All calibrations need to be performed in the inverter state, and the current needs to be calibrated at full load)

1. **Closed-loop:** Controls the closed-loop output of EG8026; **Open-loop:** Controls the open-loop output of EG8026.
2. **Output Voltage** Calibration: During normal operation, measure the actual AC voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Output Voltage Calibration" button. At this time, the calibration is not saved.
3. **Bus Voltage** Calibration: During normal operation, measure the actual bus voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Bus Voltage Calibration" button. At this time, the calibration is not saved.
4. **Resistor Current** Calibration: (This calibration value corresponds to the total current of the sampling resistor under the H-bridge arm) During normal operation, measure the actual AC current value, multiply it by 100, retain the integer part, fill it into the input box on the right, and then click the "Output Current Calibration" button. At this time, the calibration is not saved.

5. **Hall Current** Calibration: (This calibration value corresponds to the current collected by the Hall chip) During normal operation, measure the actual AC current value, multiply it by 100, retain the integer part, fill it into the input box on the right, and then click the "Hall Current Calibration" button. At this time, the calibration is not saved.

6. **Save:** After ensuring that the calibrations are correct, click the "Save" button. At this time, all calibration values are saved.

PS. For the instructions of the remaining buttons, please refer to the [EG1615 console instructions](#).

10.2.1·EG8025 Console Usage Instructions

PS2. Instructions for Using When the Serial Port is Connected to EG8025 (All calibrations need to be carried out in the inverter state, and the current needs to be calibrated at full load)

1. **Closed-loop:** Controls the closed-loop output of EG8025; **Open-loop:** Controls the open-loop output of EG8025.
2. **Output Voltage** Calibration: During normal operation, measure the actual AC voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Output Voltage Calibration" button. At this time, the calibration is not saved.
3. **Bus Voltage** Calibration: During normal operation, measure the actual bus voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Bus Voltage Calibration" button. At this time, the calibration is not saved.
4. **Resistor Current** Calibration (This calibration value corresponds to the current of the sampling resistor on the left H-bridge arm in EG8025): This calibration method is not recommended for EG8025.
5. **Hall Current** Calibration (This calibration value corresponds to the current of the sampling resistor on the right H-bridge arm in EG8025): This calibration method is not recommended for EG8025.
6. **Save:** After ensuring the calibrations are correct, click the "Save" button. At this time, all calibration values are saved.

PS. For the instructions of the remaining buttons, please refer to the [EG1615 console instructions](#).

11 Q&A

11.1 Q: How to calibrate the battery voltage? voltage?

A: See [5.1 EG1615 console](#), items 7 and 8.

A: Refer to Article 7 and 8 of the [EG1615 console in Section 10.1.](#)

11.2 Q: How to calibrate AC output voltage and output current ? the AC Output Voltage and Output Current?

A: See [5.2.1 EG8026 console](#).

A: Refer to the [EG8026 console in Section 10.2.1.](#)

11.3 Q: How do I modify the parameters?

A: 1. See [3.1 EG1615 for parameter modification.](#)

2. EG8025/EG8026 See [3.2 EG8026/EG8025 parameter modification](#)

A:

1. [For EG1615, refer to 8.1 Parameter Modification of EG1615.](#)

2. [For EG8025/EG8026, refer to 8.2 Parameter Modification of EG8026/EG8025.](#)

11.4 Q: How to calibrate the battery current? current?

A: After reading the [battery current](#) calibration value from the device information, a new calibration value is written by calculation.

The new battery current calibration value is calculated by multiplying the current current by the actual current and then dividing the current displayed by the current.

A: After reading the battery current calibration value in the device information, write in a new calibration value through calculation.

The new battery current calibration value is calculated as the current battery current calibration value multiplied by the actual current and then divided by the displayed current.

11.5 Q: How to calibrate the power or current of the EG8025 ?

the power or current of EG8025?



Figure 11-5-1

A: See steps 2-5 for the operation instructions.

1. The output current calibration value of EG8026 corresponds to the left bridge current calibration value of EG8025, and the Hall current calibration value of EG8026 corresponds to the right bridge current calibration value of EG8025. 2. Click the two "Read" buttons in step 2 respectively to read the two current calibration values, and use the read values as the current calibration values. 3. Use the AC current value displayed on the host computer (step 3) as A1, and the current value displayed on the actual power meter as A2.
 4. The new calibration value is calculated using the formula: New calibration value = Current calibration value * A2 / A1. Round the new calibration value to the nearest integer. Then fill in 4.
 5. After writing, click "Write" in step 5, and then click "Read" in step 2 to ensure successful writing.
- PS: It is recommended to turn off the inverter before performing step 5. The inverter can be turned off through the console.

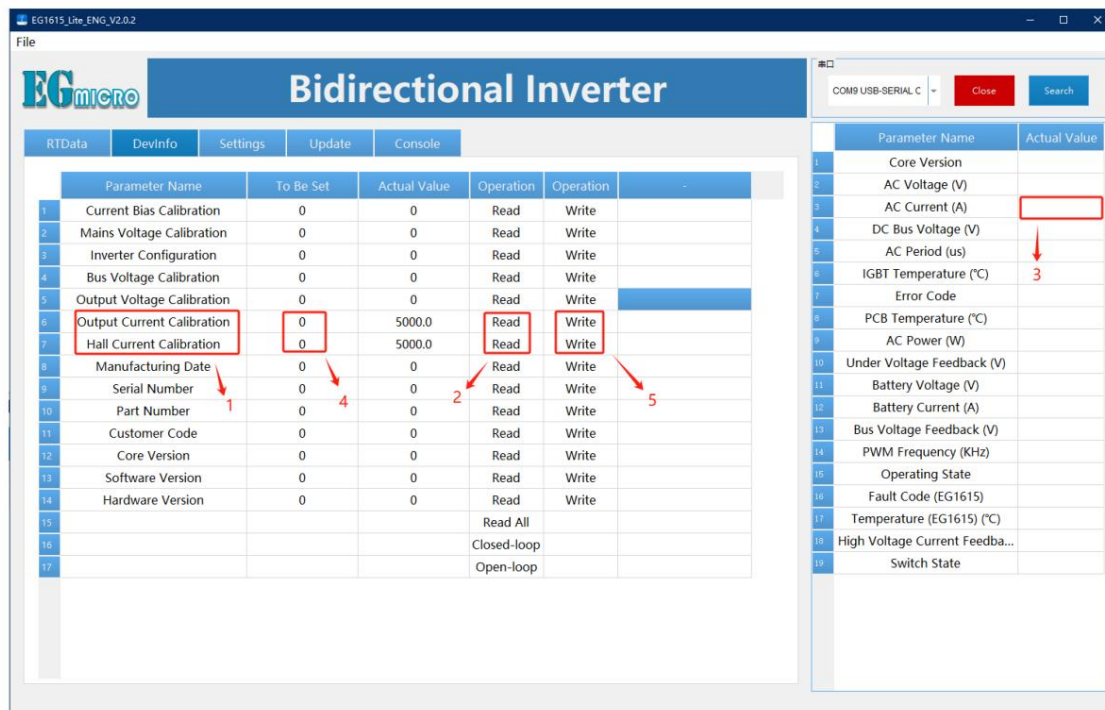


Figure 11-5-2

A: The operation steps are shown in 2 - 5.

1. The output current calibration value of EG8026 corresponds to the left bridge current calibration value of EG8025, and the Hall current calibration value of EG8026 corresponds to the right bridge current calibration value of EG8025.
2. Click the two "Read" buttons in 2 respectively to read out the two current calibration values, and the read values will be used as the current calibration values.
3. Take the AC current value displayed in the host computer 3 as A1, and the current value displayed on the actual power meter as A2.
4. The calculation formula for the new calibration value is: new calibration value = current calibration value * A2 / A1. Fill in the new calibration value after rounding it to an integer in 4.
5. Write respectively by clicking the "Write" in 5. After writing, click the "Read" in 2 again to ensure the writing is successful.

PS: It is recommended to turn off the inverter before performing step 5. The inverter can be turned off through Console – Turn Off.